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Design and implementation of an intelligent educational interaction system with integrated multimodal emotion recognition and adaptive content delivery

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Abstract
In this paper research work present a smart learning interaction system, which combines multimodal affect recognition with adaptive content selection to enable affect-sensitive and personalized learning. The proposed system recognizes the emotional condition of the learners using three major input modalities, which are facial expression, voice input and text response. Convolutional Neural Networks (CNNs) are used to recognize emotions on the face, and Mel-Frequency Cepstral Coefficients (MFCC) and Bi-directional Long Short-Term Memory (BiLSTM) networks are applied to analyse speech signals. Also, a BERT-based transformer model is used to interpret the textual inputs of the learners to obtain the contextual sentiment. These multimodal emotional cues are combined with a decision level ensemble method in order to enhance precision and trustworthiness. After identifying the emotional state, an adaptive delivery engine powered by reinforcement learning dynamically adapts the learning content, such as the adjustment of the difficulty level, format (e.g., visual, auditory, textual), and pacing to match the emotional and cognitive state of a learner. The system was piloted in a controlled e-learning environment with a sample student group. Evaluation outcomes showed significant improvement in engagement, understanding, and user satisfaction, measured through post-session surveys, quiz performance, and interaction logs compared to conventional passive learning systems. The study implies the possibility of integrating affective computing and intelligent tutoring systems to make educational technologies more humane and effective, which will move the future of human-centred digital learning systems.

Keywords Multimodal emotion recognition, Adaptive content delivery, Affective computing, Intelligent educational system, Reinforcement learning

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1 Introduction

The sphere of digital education has experienced the era of unrivalled development over the past few years due to the rapid growth of artificial intelligence, mobile technologies, and the quality of network access (Feng et al. 2025). With more and more learners surrounded by digital platforms in a variety of settings, including formal institutions and more informal online and distance education, a few fundamental questions are begging to be answered: how can we make digital learning more effective and engaging, as well as more human-centred? Although such traditional e-learning systems are effective when it comes to the scale and delivery of content, they usually lack the emotional intelligence and flexibility, which may be introduced by a human tutor (Anand et al. 2025). Such systems normally view learners as receivers of knowledge; they provide mass solutions with little or no concern to individual learners on their emotional and mental condition (Jarosz et al. 2021). Therefore, they often underperform in the aspects of motivation support, frustration casting aside and improvement of long-term retention (Saneiro et al. 2014).

In our system, Multimodal Emotion Recognition refers to the real-time detection of learners' emotional states by integrating data from facial expressions, voice tone, and physiological signals such as heart rate and skin conductance. This allows the system to accurately assess emotions like confusion, interest, or frustration. Based on these insights, the system employs Adaptive Content Delivery, which dynamically adjusts the learning material in real time—simplifying content, offering hints, or providing motivational feedback—to better suit the learner's current emotional and cognitive state, thereby enhancing engagement and understanding.

Interdisciplinary studies on the role of emotional awareness in learning have grown increasingly in affective computing, educational psychology, human-computer interactions, and several other fields (Mandava 2023). Emotions are extremely connected with cognition and their participation is important in terms of attention, memory, decision-making and motivation (Chen et al. 2025). Adverse emotional states such as boredom, anxiety, or confusion can significantly hinder learning performance, while positive emotions like curiosity, engagement, and eagerness are strongly linked to increased interest, motivation, and academic achievement. Thus, the development of emotion-aware educational systems is becoming one of the new directions of personalized learning (Saleem and Aslam 2025). The idea behind these systems is to identify in a real time the affective states of learners and change their behaviour based on the same (Heredia et al. 2022). Emotion recognition, which is multimodal by means of facial expressions, tone of voice and text entries, has been found to be especially promising in capturing profound levels of emotion, particularly once students communicate in billing or hybrid environments (Khediri et al. 2024).

In order to make an emotionally responsive system, identifying emotions is not just enough. The actual strength is in the ability to use such affective data to personalize learning in a dynamic way (Zhao and Qiu 2025). Adaptive learning systems allow changing according to particular performance, preferences and emotional response of a learner (Annapareddy et al. 2025). Such technologies as reinforcement learning methods that enable a system to learn the optimal strategies by trial and error will be especially helpful (Gupta et al. 2023). They offer a loose structure on which complex interactions

between states of the learner and the pedagogical strategies can be modelled (Zhou et al. 2024). Changes in the.

instructional content, format and pacing can be used to continuously optimize such systems and help keep learners engaged and reduce frustration in the learning process, leading to better outcome (Miao 2025). Nevertheless, combining real-time emotion recognition with adaptive delivery processes presents many issues such as modalities data fusion, privacy, and limits in responsiveness of the process without intrusiveness (Zhou and Liu 2022).

We intend to use such gaps that have to be addressed by developing an intelligent educational interaction system, which can combine multimodal emotion recognition and adaptive content delivery aspects. The system receives emotional cues of three different input channels namely facial expressions, speech pattern, and text response. The emotion recognition is done based on the Facial image using Convolutional Neural Networks (CNNs) and voice (Mel-Frequency Cepstral Coefficients (MFCC) and Bi-directional Long Short-Term Memory (BiLSTM) network. In the case of text-based sentiment, a BERT transformer model is used due to deep contextualized cognition. That emotional knowledge that is heterogeneous are combined at the decision level predicting the robustness and accuracy through an ensemble method. When the emotional condition of the learner has been established, on the foundation of using a reinforcement learning a system-based adaptive engine can adapt the educational material at any given juncture adjusting such parameters as the level of difficulty, modality (visual, auditory, or textual) and tempo. This system has been tested in a controlled e-learning platform and it has produced great results surprising the learner engagement, satisfaction and knowledge retention results. In this manner, we also introduce the next step in the creation of emotionally intelligent and pedagogically responsive learning environments.

Our approach aligns with the human-centered AI and affective computing paradigm by addressing learning performance and emotional well-being in the same breath, thus going beyond the usual content-delivery-focused e-learning systems. A state-of-the-art system that uses multimodal emotion recognition is what we propose to achieve this—facial expressions (CNNs), speech patterns, and (MFCC and BiLSTM networks) and textual responses processed with a BERT model. The heart of our innovation is a robust content delivery engine configured for adaptability, that progresses past static tutoring employing Reinforcement Learning (RL) as the main teaching approach. This engine then changes instructional elements like difficulty level, content pacing, and presentation mode (visual, auditory, or textual) during the actual teaching session, according to the learner's needs, dynamically and intelligently. The proof obtained in this paper shows the significant gains that are made over learner engagement, satisfaction, and knowledge retention in comparison to the non-adaptive systems in the traditional sense, through the integrated approach. In the end, it can be said that this work, apart from proving the feasibility of building truly emotionally intelligent tutoring systems, also points out the considerable educational value that has to be derived from such systems.

2 Literature review

The research on adding emotion recognition to the smart education systems is an issue of growing interest among researchers as it may lead to individualization of the learning process and increase user interaction. Several authors have studied multimodal

detection of affect employing facial expression, speech and platforms and other systems have studied adaptive learning processes in response to the emotional affective and cognitive states of learners. The current literature evidence illustrates a high level of development that takes place in each particular modality and adaptive framework, yet there is a gap within finding many solutions on how to offer a seamless, real-time, and robust integration of these mechanisms. Other methods have some degree of sensitivity to noise or limited conditions in that they are highly dependent on one modality and thus prone to misclassification. Others utilize rule-based systems of adaptation which are non-scalable and have no power of learning. It is possible to provide a comparative overview of recent studies as Table 1 highlights the main methods of study used, their advantages, and limitations.

2.1 Recent emotion-aware learning systems

Salloum et al. (2025) suggested an emotion recognition system by AI based on the detection of emotions not through facial and emotional expressions but using analyzed and processed text without general teachers adapting teaching strategies to real-time changes. their system allows individualized learning by adapting the delivery of contents according to noticed emotions, which increases the engagement of students. Nevertheless, personalization is rather superficial without deep learner profiling or adapting in the long-term.

Wang et al. (2025) have established a closed-loop emotion regulation system with multimodal affective computing and Emotional Markov Chain model. In this method, there is proper tracking of emotions and change of feedback in the course of the learning sessions. Although the system has a lot of advantages in the area of emotional modelling,

Table 1 Comparative review of recent Emotion-Aware educational system

Author(s)	Techniques	Advantages	Disadvantages	Limitations
Salloum et al. (2025)	AI-based emotion detection (facial, text) + adaptive teaching	Real-time feedback; adaptive instruction. Improves learner engagement and adjusts pacing to individual needs.	Limited personalization depth. May not fully adapt to complex learner profiles or evolving preferences.	Lacks integration with physiological signals for a fuller emotional profile.
Wang et al. (2025)	Multimodal affective computing + Emotional Markov Chain	Accurate emotion tracking; feedback loop. Enables timely intervention to maintain learner motivation.	Complex architecture; context-sensitive. Requires extensive calibration to different learning contexts.	High computational demand may hinder real-time scalability.
Zhao & Zhou (2024)	IoT + AI-based voice & visual emotion detection	Reduces stress; interactive. Encourages a safe learning space where learners feel more supported.	IoT dependency; limited learning focus. Reliance on sensors may shift attention from core learning outcomes.	Data privacy and network reliability are critical constraints in real-world use.
Alzubi et al. (2025)	Facial, voice, and gesture-based HCI	Engaging interaction; multimodal input. Supports diverse learning styles through visual, auditory, and kinesthetic cues.	Gesture errors; less scalable. Inconsistent input recognition can affect usability in large-scale deployments.	Limited support for content adaptation based on emotion type or intensity.
Mohammed Aly (2025)	CNN for facial recognition	High accuracy; real-time monitoring. Allows for precise detection of emotional shifts during instruction.	Lacks adaptive content delivery. While monitoring is strong, content adjustment based on emotions may be limited.	Focuses only on facial cues; ignores voice and context-based emotional inputs.

the complexity of the architecture of the system and sensitivity to context make implementation challenging.

Zhao and Zhou (2024) examined the employment of IoT-based robots as entertainment devices fitted with AI-enhanced voice and vision recognition functions to promote the mental health of the students. The system is effective to reduce the stress levels and improve interactivity in schools. Nonetheless, it heavily depends on the IoT infrastructure and, therefore, is not massively applicable to academic systems due to the largely insufficient focus on learning outcomes.

Alzubi et al. (2025) designed a multimodal human-computer interaction (HCI) frameworks into smart learning environments utilizing the facial, voice, and gestural input as the source of emotion recognition. Their system allows stimulating feedback on learning and encourages interaction. However, there are a chance of error in gesture-based type inputs and too much of it cannot be scaled up hence it is problematic in remote or large level applications.

The real-time monitoring of the students with the application of high-end Convolutional Neural Networks (CNNs) to identify facial expression was the topic of Mohammed Aly (2025). The high precision of the system in tracing the learner's mood in online sessions is of assistance in the evaluation progress. But it does not have the mechanisms of adjustment of content according to the identified emotional effects, which reduces its motions on individual training.

The role of AI in education is to create more interactive and individual learning environments. Ramalingam et al. (2024) have performed an extensive review that indicates among the other advanced technologies AI-based Adaptive Learning Systems with the ability to adjust the teaching materials according to each student are the leading ones. The current research utilizes this technology and innovatively develops it by suggesting a new method that through connecting the use of emotion detection by different means expands the tenets of adaptive learning.

The primary goals of today's e-learning systems are to accurately assess student engagement, as it determines the suitability of adaptive interventions. The use of deep learning models for student engagement detection has gained traction and credibility with the work of Wang and Nandi (2024). Our proposed system not only provides unimodal and simple multimodal approaches but also offers a more thorough and robust evaluation of student engagement through the combined assessments from visual, vocal, and textual modalities. This, in turn, facilitates the delivery of personalized content in a more nuanced and timely manner.

Zhang et al. (2024) They have shown the effectiveness of deep learning methods in highly specialized educational areas, such as ideological and political education. This is a significant step in the direction of using deep learning in education to increase the scope of the spectrum methodology of the new generation, which is only limited by the structured learning environments. Our research is a continuation of this work, where we apply deep learning to the development of an intelligent educational interaction system that will combine multimodal emotion detection with adaptive content delivery, thus taking care of the most important factor of all in digital learning platforms—emotional responsiveness. Scalable and intelligent educational systems will depend a lot on powerful cloud computing and data management systems. One of the supports for this type of technology is research, mostly founded on Chen and Kumar (2023) who are at the

forefront of integrating cloud computing with big data for fault detection and security. Our project, which is structured upon such infrastructural advancements, takes these principles and puts them into practice by developing a multimodal emotion recognition and adaptive learning system that processes sensitive educational data in a secure and reliable manner within a cloud-enabled environment thereby allowing for real-time processing. The collaboration of artificial intelligence and cloud infrastructure has been recognized as one of the main factors that power the development of modern educational technologies. An example of this is found in Ganesan (2021), who presented a successful AI platform for intelligent education management powered by the cloud. Our study continues this trend but selectively puts attention on the essential yet frequently neglected aspect of emotional intelligence. Ganesan's research opened the way for AI-operated education solutions, but our platform brings along a complex system of multiple channels for emotion detection and a reinforcement learning-assisted adjustable engine to form a more 'whole', 'fast' and 'humanized' teaching/learning atmosphere.

2.2 Problem formulation and motivation of the research

Although there have been enormous improvements with regard to emotion-aware educational technologies, the studied literature has severe constraints, which limit the general usefulness of such technologies in adaptive education contexts. Some of the systems are concerned with one or a few modalities and thus they are sensitivity to context variability and noise. The rest of them do not provide powerful adaptation affordances and instead respond statically or by rule instead of making their responses based on entirely personal content delivery. Multitiered architectures will pose scalability and implementation issues (architecture with emotional Markov Chain or IoT dependency) when faced with reality of education. Furthermore, most methods lack a closed feedback loop, where the process of learning includes the emotion recognition and adapting the content to the learner, and this is crucial in ensuring that the long-term engagement and satisfaction rates of the learners are high. To resolve these shortcomings, the developed system embraces a multimodal emotion recognition model– a combination of facial expressions, voice input, and textual feedback blended with a final step of an ensemble at the decision level. This emotional understanding is then connected directly to an adaptive learning engine driven by reinforcement learning, real-time, highly personalized content modifications that react to both emotional and cognitive state of learners.

3 Proposed system model

The system that is proposed is a holistic, emotion aware smart learning in which multimodal emotion recognition and reinforcement learning based adaptive content delivery is combined to provide an emotionally intelligent and personally relevant learning environment. It starts with gathering multimodal data in the form of facial expression, voice cues, and textual answers, which provide unique information on the emotional and cognitive level of a learner. The total number of image samples was 1,200 and text samples was 1,000, which were gathered for the purpose of training, validating and evaluating the models proposed in an effective way. These inputs are used to run through specific deep learning architecture- visually, CNN, speech emotion detection MFCC with BiLSTM, and background contextual sentiment analysis of text using a BERT model. The result produced by each modality is then combined by a decision-level ensemble technique

like weighted voting and a coherent and multi-resistant emotional state classification is obtained. This process of fusion makes the ultimate emotion deemed to be non-noisy even though the individual modalities might be noisy or even inconsistent. The proposed architecture can be found in Fig. 1.

As soon as the emotional state of the learner is determined, the system develops a dynamic profile of the learner in which cognitive measures such as the level of engagement, quiz results and behavioural trends are taken into account. A RL agent utilizes this profile, as the central point of the adaptive content delivery engine. The RL agent can understand the current affective-cognitive state of the learner and choose effective instructional actions, which could be a change in the level of the content, the speed at which an instruction is covered, or the format type of an instruction (visual, auditory, or textual). In the process, a real-time feedback loop changes the learner state making the profile update in real time, so that the system has the ability to change every moment.

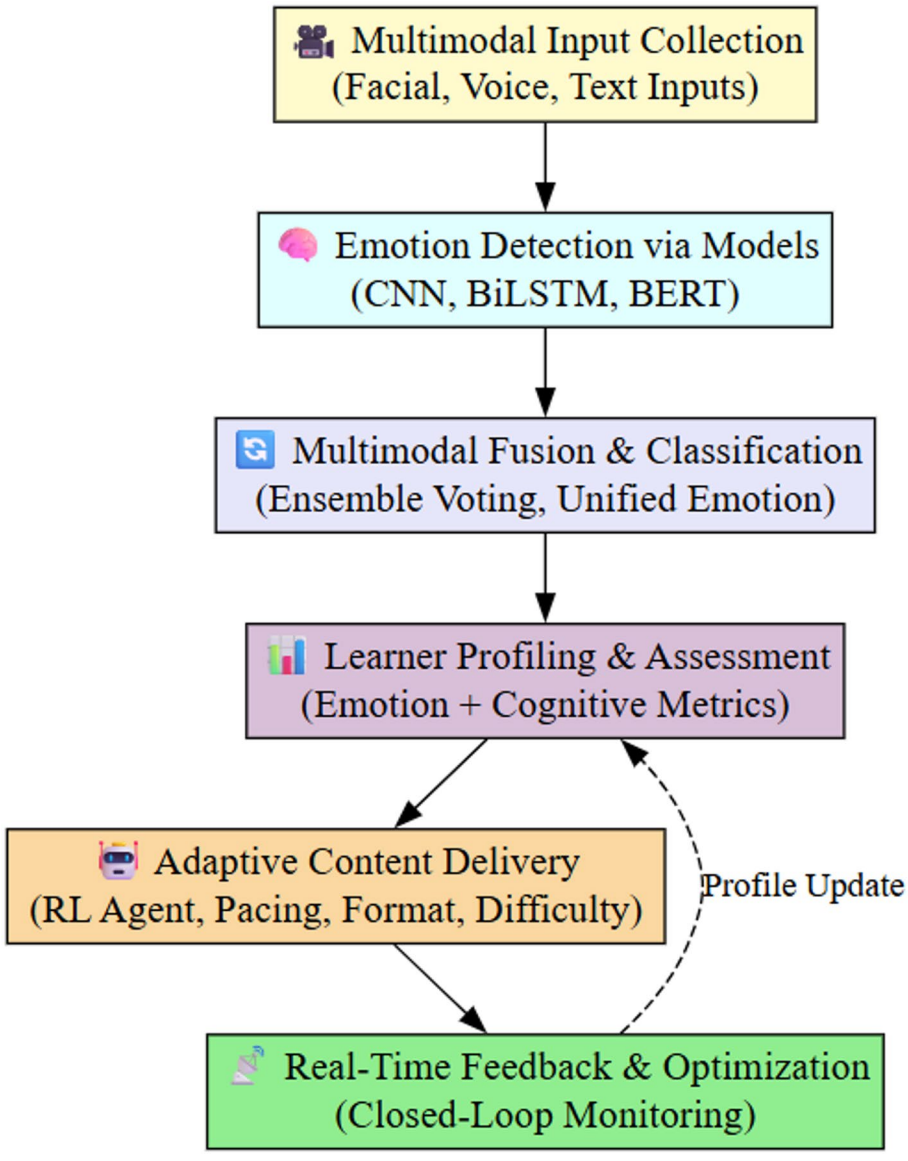

Fig. 1 Proposed Design Architecture

This closed-loop design would make learning personal, emotionally supportive and pedagogically effective, and is one forward-looking answer to an emotionally responsive digital education.

3.1 Multimodal input collection

A holistic method of data collection must be implemented in the foundations of affect-sensitive learning system because precisely recording the emotional condition of learners is necessary. This is actualized by the use of multimodal input collection that combines three major user feedback including facial expressions, voice data and text feedback. These modalities provide diverse and complementing clues to emotional and cognitive status of the learner. There is a facial identification via a webcam that gives a high amount of visual feedback (e.g., smiling/frowning/eyebrows up/eyes slit) and correlates to a certain emotional reaction (e.g., interest, confusion, boredom). At the same time, the speech is documented using a microphone to document such vocal characteristics as pitch, tone, volume and rhythm. These are the auditory cues that tend to be unconscious and point to the inner moods of stress, excitement, or frustration that are unlikely to be evident on face alone in the first place (Pordoy et al. 2024).

The third essential modality is represented by textual forms of input (e.g. chat messages, written responses, or typed comments). The input represents the linguistic and semantic content of communication of learner and usually contains references to emotional sub-current or manifestations of emotions and knowledge. All three modalities allow creating a multidimensional emotional data vector of individual learners. Mathematically this multimodal input at time 260 t can be depicted as Eqs. (1),

$$E_t = [F_t, V_t, T_t] \tag{1}$$

where F_t features obtained based on facial expression data, V_t features made based on voice input, and T_t features based on the input of textual data. Such a combined vector E_t is fed to the next emotion recognition models so that the affective state of the learner can be understood in a synchronized and comprehensive manner. By combining these various signals, one is able to alleviate shortcoming of any individual modality in order to be more robust and enhance reliable detection of emotion across the various learning conditions (Savchenko et al. 2022).

3.1.1 Datasets and input specifications

The creation and approval of the multimodal emotion recognition system took the three benchmark datasets that were already established as reference points and picked each one according to its applicability n the educational settings. The FER-2013 dataset was the one used for the facial expressions analysis and it contains 35,887 grayscale facial images (48 × 48 pixels) categorized into the seven main categories of emotions. Speech emotion detection was done through the RAVDESS dataset, which is composed of 7,356 audio samples with 24 professional actors performing the eight basic emotions at two different intensity levels, thus allowing a thorough variation of the vocal affect to be modelled. The Go Emotions dataset came next for the textual sentiment analysis and it includes 58,000 Reddit comments that had been carefully selected and tagged with 27 different emotions, thus giving a deep and rich understanding of the learners' reactions in their contexts. The splitting of the datasets into the standard 80-10-10 for training,

validation, and testing took place in all cases, and the application of the suitable data augmentation techniques to counterbalance class imbalance and promote model generalization over different learner groups and environmental conditions was also adopted.

3.2 Emotion detection via specialized models

As soon as the multimodal emotional data is gathered, it is passed to individual modality data through a specified deep learning model that has been developed to consume emotion-specific feature. CNNs process facial expression and they are very useful in recognizing spatial structures in the visual images. CNN model draws in the image of face where a filter is applied to the facial image that has been pre-processed, which gives layers of filters to identify key features on the face like moving eyes, lips on the face to curl and brow position. Then they are categorized into the simple emotional states (e.g. happy, sad, confused, bored), giving the system the probability of guessing how the learner expresses himself or herself in ways that are visually apparent of the current mood (Villegas-Ch et al. 2023).

At the same time, the voice input process is realized in two phases. Short-term power spectrum of sound is then extracted as MFCCs which identify important acoustic characteristics of speech. These MFCCs then are submitted into a BiLSTM network, which is used to learn the temporal dependencies of sequential data. BiLSTM model gives the model the advantage of both previous and upcoming context of speech pattern to enable the model to recognize emotional tones of speech such as anxiety, calmness, or passion.

In the case of textual response, a transformer model is used which is based on BERT to find contextual sentiment. BERT can read not only the meaning of single words but also their context in a sentence and is therefore perfect in making subtle emotional judgment based on the written text of learners. Emotion detection process in the three modalities can be generalized as Eqs. (2),

$$\widehat{e}_m = f_m\left(x_m\right) \quad for\ m\ ?\{F, V, T\} \tag{2}$$

Here, x_m representing the output of the modality-specific model m (Facial - F, Voice - V, Text - T), $\widehat{e}_m$ the predicted emotion label on that modality corresponding to the pre-trained models. These are the independent predictions and are subsequently combined into one coherent emotional knowledge and provide the foundation of adaptive learning interventions (Fernández Herrero et al. 2023).

3.2.1 Mathematical formulation of modality-specific models

CNN architecture for facial recognition: The CNN has a multitude of feature detectors in various layers that take out hierarchical patterns from face images sequentially. The layer carries out convolutions as well as nonlinear activations and spatial pooling, and by the end of the process, the initially raw pixel values have been converted to high-level emotive feature representations through a series of weighing kernel operations and matrix transformations.

MFCC feature extraction for speech processing: The extraction of Mel-frequency cepstral coefficients converts raw audio signals to spectral features by first computing the power spectrum, then passing it through a Mel-scale filter bank meant to imitate the human auditory system, followed by logarithmic compression and discrete cosine

transformation thus yielding cepstral coefficients that are uncorrelated and that represent the characteristics of the vocal emotion.

BiLSTM architecture for temporal modeling: The bidirectional Long Short-Term Memory network takes sequential speech features through the component of gated memory cells parasitically and thus, it very judiciously retains and combines information from the different time steps. The design is exceptional as it has both the forward and backward temporal contexts which also help the model to detect long range dependencies in speech patterns through very intricate gate mechanisms and state changes throughout the sequence.

3.3 Multimodal fusion and emotion classification

The next important step after receiving individual predictions of emotions based on facial, voice, and textual signals using their dedicated models is to merge their results into one and the same coherent emotion prediction. This is required as every single modality will record varied dimensions of the emotional status of the learner- facial expressions can demonstrate an exterior effect, voice tone may show inner tension or absorption and text can express either an emotional nature or intent (Sowjanya and Krithiga 2024).

Nevertheless, using one modality only may become inaccurate because of the presence of noise, short-term distractions or unclear signals. As an example, a learner would have a neutral facial expression when he or she is being frustrated in his or her tone or text. Therefore, multimodal fusion is quite a reliable and robust solution because it takes into account all available emotional signals (Jagadeesh and Baranidharan 2022).

The process of fusion is usually done at the decision level also referred to as late fusion, where models put out an estimate of emotion label as well as a confidence score by each modality. More predictions are then combined based on an ensemble learning strategy (e.g. majority voting, weighted voting or stacking). The resultant emotion is the one that all the modalities predict in the case of the majority voting (AlZu'bi et al. 2022). In weighted votes a more accurate/ more reliable modality takes more claim in the ultimate resolution, depending on the particular situation or the past Performance Quality (PQ). To use another example, let us assume that the model of emotion that relies on speech is more likely to be more accurate in noisy settings than the one based on facial recognition, then it can be given a greater weight. Mathematically simple weighted decision fusion can be expressed as Eqs. (3),

$$\widehat{E} = \underset{e}{\operatorname{argmax}} \sum {}_{m=1}^{M} w_m . \mathbb{I}\left(\widehat{e}_m = e\right) \tag{3}$$

In this case, M represents the number of modalities, $\mathbb{I}(.)$ is the indicator function that delivers 1 when the condition is fulfilled and 0 otherwise, w_m is the weight given to modality m, the predicted state of emotion $\widehat{e}_m$ is the value e m and the final predicted emotional state is $\widehat{E}$.

Justification for decision-level fusion approach

The decision to use the decision-level fusion (late fusion) method instead of feature-level or model-level methods was based on several reasons that were strategically important for the educational sector. Firstly, it recognizes the different nature of our multimodal inputs—images, audio waveforms, and text tokens, all requiring separate,

heavy-duty deep learning architectures for the best feature extraction. Secondly, this method gives very important operational flexibility; if one of the modalities goes off (for instance, if microphone permission is denied, or the student turns off their camera), the system can still operate with the remaining modalities, thus degrading in a nice way rather than stopping completely. Lastly, at the decision-level, the context-aware, weighted voting can be applied, where the confidence of each modality experts can be dynamically modified as per the real-time signal condition (for instance, in less lighting conditions, the weightage of facial analysis can be reduced), thus, improving the overall strength and dependability of the system in the real-life, ever-changing learning environment.

3.4 Learner profiling and state assessment

Having correctly categorized the emotional state based on multimodal fusion, the system can then be used to create a dynamic profile of the learner, which combines emotional and cognitive aspects. This profile is the focal point to the learning experience of every learner (Munshi et al. 2020). This is in contrast to the static profile of the user which at best can be based on either demographics or past performance and at worst on little or no data at all since it is not updated as the learner interacts with the system over time. The emotional responses to facial expression, tone of voice and sentiment of the text are captured along with the performance data relating to the correctness of quiz answers, time of response and behaviour engagement (e.g. the number of interactions, pause behaviour and time on the task).

This allows this system to follow the trends, so to speak, of emotion through time, e.g., determine whether a particular learner tends to exhibit frequent displays of frustration when processing specific content forms or subject matter. In the same vein, the signs of motivation, curiosity, or boredom may be placed on a graph with the levels of difficulty and teaching strategies. Cognitive load may also be measured by examining the rate at which a learner is assimilating information or even whether he/she needs exposures to be able to understand some concepts. The insights are applied to name the current affective-cognitive status of the learner, including such labels as being highly engaged and confident, confused but motivated, or disengaged and overwhelmed (Zhu et al. 2022).

This robust, time-based learner profile allows the system to make sound and specific judgments concerning the ways of tailoring the learning experience. It enables context-sensitive personalization of the past history of individual behaviour and affective reactions to be taken into account as criteria on which future content is chosen, timed and possible support processes. This way, the platform is not only reactive to temporary feelings but it is aware and responsive to wider learning paths and as a result the learning environment is more facilitative and functional (Pabba and Kumar 2022).

Comprehensive assessment of learner profiling parameters

The process of profiling the learner in our case takes into account not only emotional but also cognitive metrics that together yield a complete state evaluation. The emotional metrics are the following: (1) the Engagement Index - which is based on the amount of time the subject was looking at the screen and the number of positive expressions; (2) the Frustration Level - which is estimated from the density of brow furrow, variance of speech pitch, and negative sentiment keywords; (3) the Confidence Score - which is evaluated by the latency of response and the fluency of verbal patterns. The cognitive

metrics are: (1) Knowledge Retention Rate - keeping track of the decay of quiz performance over the course of sessions; (2) Concept Grasp Speed - measuring the time-to-mastery for new subjects; (3) Learning Pathway Efficiency - looking at the best content sequences for individual learners. Our reinforcement learning framework perpetually updates these metrics and runs correlations among them, which allows the system to recognize temporary emotional states as well as persistent cognitive patterns and thus deliver truly individual educational interventions.

3.5 Adaptive content delivery using reinforcement learning

The most important aspect of personalized learning is dynamically adjusting learning content to the needs of the individuals. This is done by the use of an adaptive delivery engine, which uses RL to make intelligent decisions on content adjustments to provide to the learner, and is based on the constantly changing learner profile (Razzaq et al. 2023). Reinforcement learning A form of machine learning in which, using a reward signal, an agent tries actions in its environment and improves its understanding of their desirability. In this, the RL agent senses the current state of the learner in terms of the emotional-cognitive state and decides which learning action (factors such as the degree of difficulty of a given question, the medium used in the explanation (visual, auditory, textual) or rate of content delivery) is best suited (Gazawy et al. 2023).

Every step of the RL agent will have an impact on the involvement of the learner, his performance and an emotional response' which would be used to update the reward signal. In other words, when the learner seems to be more active and to have better results after changing textual to visual content, the agent gets a positive reward, and it understands that it should choose such actions in the next situations like this. The system will in time be more successful in determining what kinds of content and approaches will have the greatest impact on learning in any particular person (Lin et al. 2020). The agent aims at maximizing a cumulative reward, which represents long-term educational success that might be the long-term engagement, greater understanding and emotional feeling. Mathematically RL framework is considered as a Markov Decision Process (MDP) in Eqs. (4),

$$< S, A, P, R, \gamma > \tag{4}$$

In this case, S is the pair of the learner states (cognitive and emotional profiles), A is the set of possible adaptive choices (pacing, format, content type, etc.), P is defined the transition probability between states, R is defined the reward function depending on the learner responses/ feedback, and the discount factor γ is defined as a balance between current and longer gains (Garcia-Garcia et al. 2023).

3.5.1 RL agent, pacing and difficulty functions

At the heart of our adaptive engine is a Deep Q-Network (DQN) agent. The state space S, which is at the heart of our adaptive engine, explicitly includes the emotional-cognitive profile of the learner as described in Sect. 3.4. The action space A is tailored with delicate controls for:

Difficulty Adjustment: D(s) function that simulates problem complexity change in real time through Item Response Theory, where the difficulty parameters are altered along with the emotional involvement (frustration lowers difficulty, confidence increases it).

Pacing Control: A policy π_pace(s) that manages content flow using a Markov-modulated Poisson process, where the transitions between slow/medium/fast delivery are controlled by confusion states and cognitive load metrics.

The reward function R(s, a) includes the following:

$$\mathrm{R_engagement} = \lambda_1\,(\mathrm{time_on_task}) + \lambda_2\,(\mathrm{positive_affect_score})$$

$$\mathrm{R_performance} = \lambda_3\,(\mathrm{quiz_accuracy}) + \lambda_4\,(\mathrm{concept_mastery_rate})$$

$$\mathrm{R_efficiency} = \lambda_5\,(\mathrm{content_coverage}) - \lambda_6\,(\mathrm{cognitive_overload_indicator})$$

By this setup, the reinforcement learning agent is in the position to learn the most useful teaching methods, considering emotions as a factor.

3.5.2 Real-time feedback and continuous optimization

During this end-state, the system can make use of a real time feedback platform to keep tracking on the emotional state and performance of the learner all through the session. Using such live signals as facial expression, tone of voice, sentiment analysis of textual posts, and behaviour pattern (e.g., accuracy, response speed, lingering) the system detects changes of emotion and a cognitive effect that might be signified by engagement, confusion, or frustration. This information is fed back into a closed loop control system that allows real time and dynamic control of the content delivery: varying the format, speed or difficulty level to better match the needs of the learner. An example would be that; disengagement signs can cause the system to change into visual explanation and mastery will bring in more advanced issues. This ongoing process guarantees that the learning process is made emotionally sustaining, cognitively suitable, and educationally effective leading to a greater level of involvement and better learning in the long term (Leong 2020). In addition to its adaptive capabilities, the system must also address key ethical considerations. These include ensuring data privacy and informed consent when collecting emotional and behavioural data from learners. It is essential to maintain transparency in how emotional data is used and to prevent potential misuse or emotional manipulation. Ethical safeguards help protect learner autonomy and foster trust in emotion-aware learning technologies.

3.6 Multimodal algorithm framework

Pseudo code

```
The integrated algorithmic approach can be summarized as:

Input: Facial frames F, voice signals V, text inputs T
Output: Adaptive content actions A

1. Feature Extraction:
  - Facial: CNN(F) → e_f
  - Voice: MFCC + BiLSTM(V) → e_v
  - Text: BERT(T) → e_t

2. Multimodal Fusion:
  E_fused = WeightedVote (e_f, e_v, e_t)

3. Adaptation Decision:
  A = RL_Agent(E_fused, learner profile)
```

This research implements a structured algorithmic framework where facial expressions are processed via CNN, speech signals through MFCC-BiLSTM pipelines, and textual inputs using BERT transformers. These multimodal outputs undergo decision-level fusion using weighted confidence scoring, with the consolidated emotional state informing a reinforcement learning agent that dynamically optimizes content delivery through continuous reward-based policy updates.

4 Outcome and evaluation

The section introduces inputs and discussion of the planned smart interaction system of learning that combines both multimodal affect recognition and reinforcement learning applied adaptive content. This program was piloted in a regulated e-learning setting in order that it might assess the success of the program in bringing together learner involvement, emotional correlation and scholastic achievement than conventional non-adjustive learning frameworks. Level of accuracy of emotion identification, pre and post-test (learning gain), engagement measures (time-on-task, frequency of interaction) were obtained as a quantitative measure, alongside qualitative comments pertaining to user satisfaction. It has proven that the system can be used effectively due to the results, as the adaptive approach to changing a product based on emotional responsiveness greatly improved emotional responsiveness, eliminated learner frustration, and facilitated overall motivation. The findings also demonstrate the resilience of the fusion model employed to classify the emotions as well as the versatility of the RL-based content delivery mechanism developed. The results support the prospects of affect-sensitive systems to transform electronic learning with real-time human cantered instructional environments.

The confusion matrix Fig. 2 shows how effective the emotion classification model can be based on five emotion categories of Happy, Sad, Angry, Neutral, and Surprised. The model returned the highest accuracy of Surprised class with an accuracy of 0.92 followed by placement of Happy at 0.91, Angry at 0.89, Sad at 0.88 and Neutral at 0.86, which indicates that the model is reliable even when coming to different emotions. The misclassifications were not very substantial; Sad was confused 5% of the time with Angry and Angry was classified 4% of the time as Sad and Neutral. Moreover, 3% of the Neutral

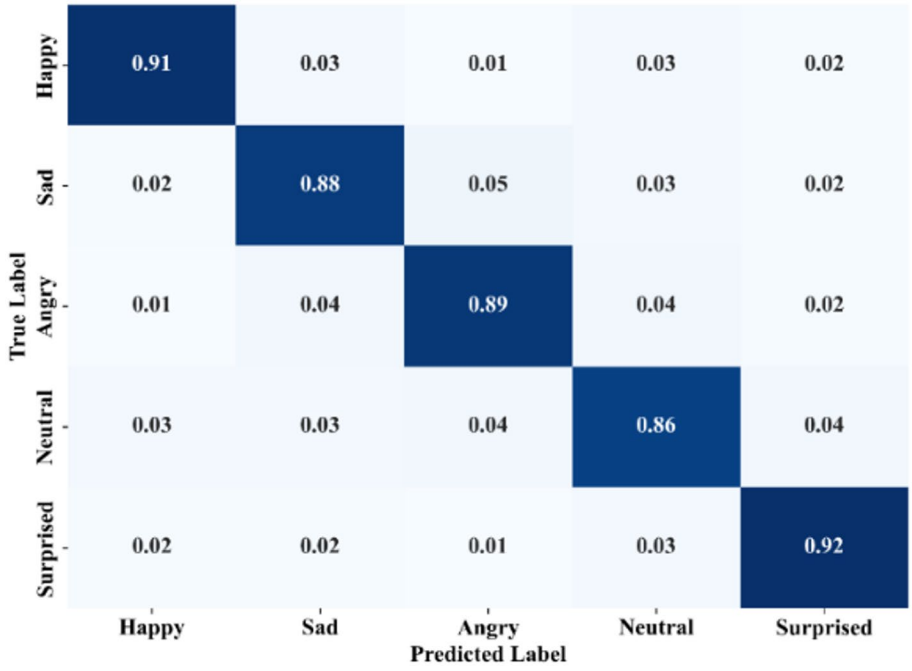

Fig. 2 Confusion Matrix

labels were confused with Happy and Sad, and the Surprised label was associated with few misrecognitions, only 2% being confused with either Happy or Sad. These findings emphasize the effectiveness and precision of the multimodal emotion recognition methodology proving its appropriateness concerning detectable emotions despite adaptive learning sites. These confusions are due to overlapping facial expressions and similar behavioural cues among emotions. Such nuances highlight the need for robust multimodal data to improve emotion differentiation in adaptive learning environments.

The performance of the various emotion recognition model (Face (CNN), Audio (BiLSTM) and Text (BERT) and Proposed Fusion) are compared on Accuracy, Precision, Recall, and F1-Score basis as shown by the Fig. 3. The individual modalities were the Audio (BiLSTM) and Text (BERT). Both models demonstrated similar, but the former was slightly better with an accuracy and recall of about 0.76 and 0.77 respectively whereas the other achieved a slightly higher precision and F1-score of 0.78 and 0.76. The recall of such Face (CNN) model was the lowest (~0.72), which means that it had the weakest ability to recognize all applicable emotional patterns. However, the Proposed Fusion model has scored higher that all single-modality methods in all metrics scoring due to an approximate 0.91 in accuracy as well as recall, 0.89 in precision and 0.90 in F1-score. This shows the effectiveness of the decision-level fusion technique that optimally integrates the complementary features of every modality producing much more accurate, stable and reliable recognition of emotion in an adaptive learning setting.

Figure 4 shows the proportion of emotion types used in the dataset, which is characterized by the amount of the samples that can be used per a type of feeling. The Neutral category of emotions among the seven categories has the most samples of 1400, the categories second in the number of samples are Happy and Sad with 1200 and 950 samples respectively. The percentage of the Angry is 870 samples and Surprised is 760. Fear with 620 samples and Disgust with 580 samples are the two least represented classes in the

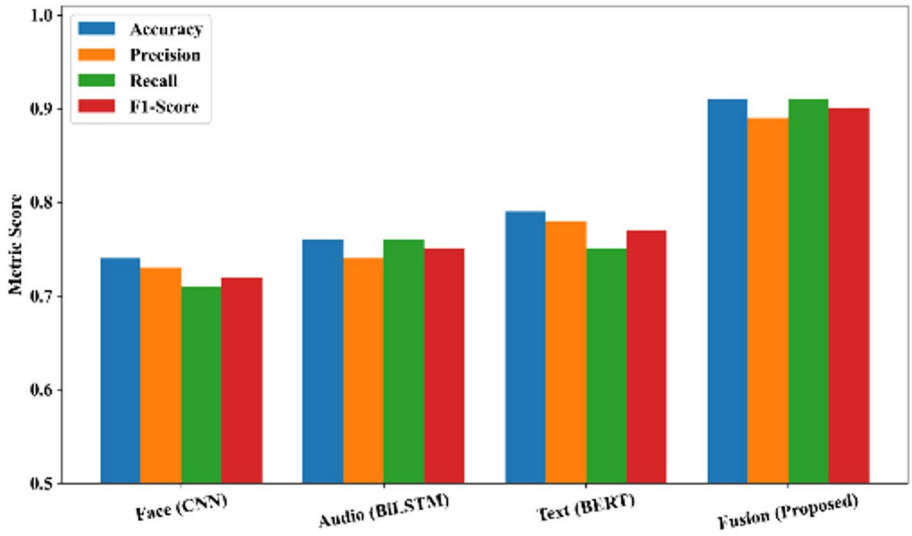

Fig. 3 Performance measure

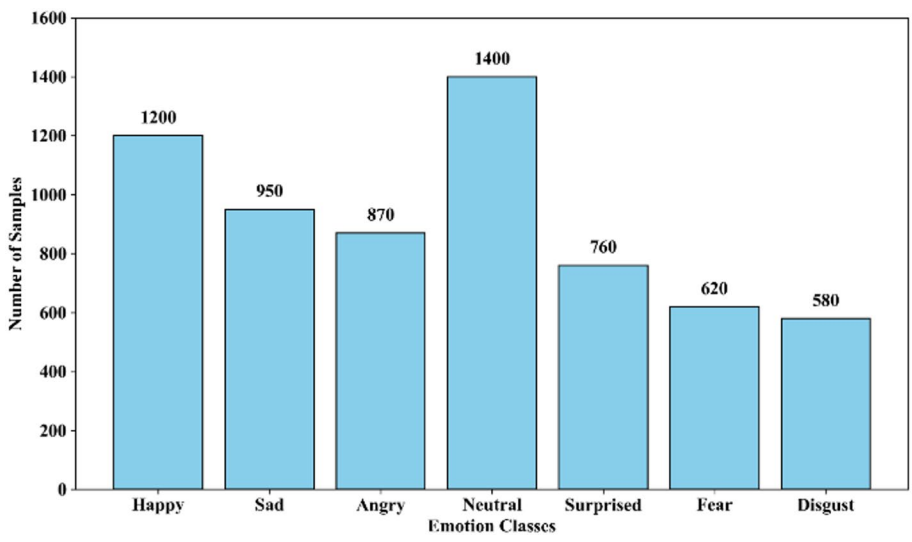

Fig. 4 Emotion class analysis

data. This shows that the dataset has a moderate imbalance with the predominance of Neutral and Happy categories and could affect the learning behaviour and bias of the model. It is preferable to achieve a balanced representation or use appropriate re-sampling or weighting methods to make the classification fair and prevent overfitting to the large classes when training.

Reliability Fig. 5 shows calibration performance of the proposed emotion recognition model with some plot points of true accuracy against mean predicted confidence and the dotted line indicates the equal to zero value. The dashed diagonal line indicates perfect calibration in which the estimated confidence equals the actual accuracy. The blue spot describes the calibration of the suggested model. In higher levels of confidences (0.6 to 1.0), which is where the model generally follows the perfect calibration line accurately, confidence counts and actual accuracy are still very high above 0.7; representing high dependability of any model at the profitable levels of prediction. Nevertheless, a

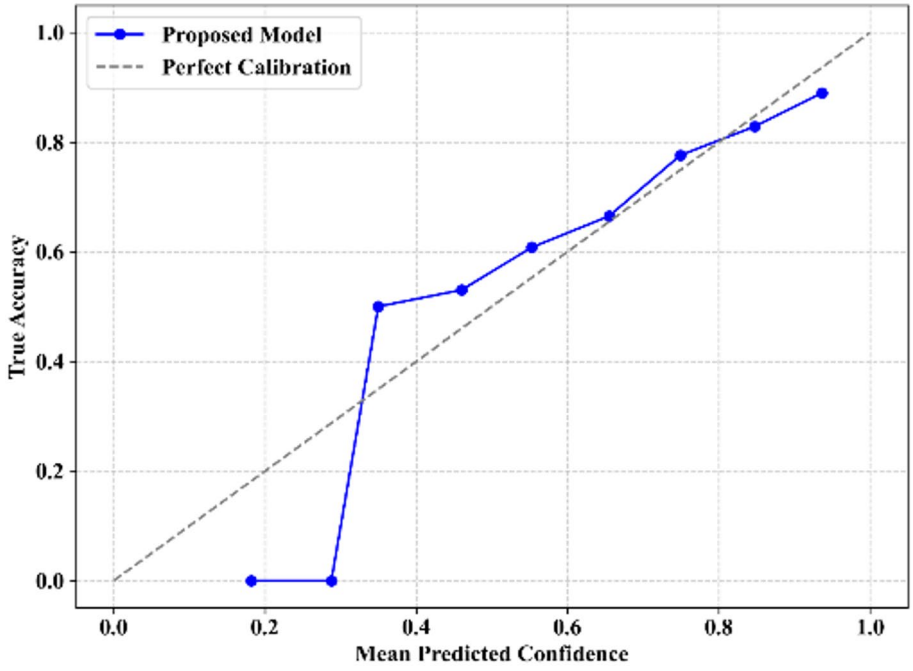

Fig. 5 Proposed model calibration

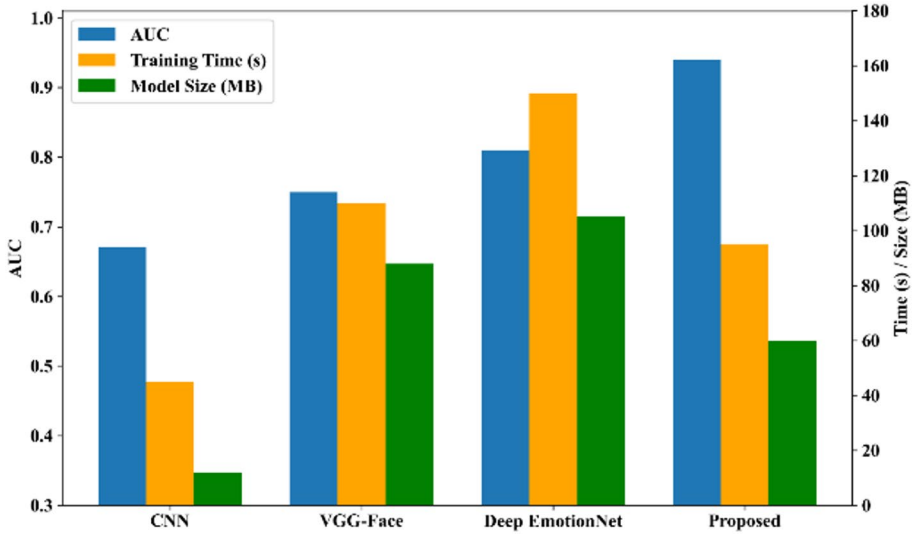

Fig. 6 Emotion model AUC validation

few outliers can be observed within the lower confidence interval (between 0.2 and 0.4), which implies that the model under-predicts its performance, so the results are rather conservative. A well-calibrated behaviour of the model is observed in general, including in the higher confidence regions affecting the decision-making process, despite minor discrepancies in zero-confidence regions, which confirms its appropriateness in emotionally sensitive and dynamic learning environments, where a high degree of trust is placed on the model in probabilistic assertions.

The Fig. 6 shows a comparison of 4 models in emotion recognition, CNN, VGG-Face, Deep EmotionNet and Proposed in regard to their performance in three aspects which

are AUC (Area Under the Curve), training time (seconds) and model size (MB). The Proposed model has the best (almost 0.95) AUC, which means that it classifies better than Deep EmotionNet (CNN) (slightly more than 0.81), VGG-Face (around 0.75), and CNN (around 0.67). Regarding the time spent on training, Deep EmotionNet takes the longest (~150s), followed by VGG-Face (~120s) and the Proposed model (~95s), and CNN trains the fastest (~50s). Concerning the model size, Deep EmotionNet model is the heaviest (~110 Mb), and VGG-Face is the heaviest (~95 Mb), whereas the Proposed model represents a compromise with middling size (~65 Mb), which is considerably larger than CNN (~35 Mb) but less resource-intensive than the deeper ones. Such findings underscore the performance of the Proposed model, which not only has the highest accuracy, but also takes an acceptable training time and size, so it is good to be deployed in the real world with a scalable application of the affect-aware learning systems.

Figure 7 provides a comparative analysis of four emotion recognition models namely, CNN, VGG-Face, Deep EmotionNet, and Proposed model according to three important performance measure namely Generalization Accuracy, Adaptation Success Rate, and Ablation Impact (%). The proposed model outperforms all other models by demonstrating superior generalization across diverse learning situations. It achieves the highest generalization accuracy (~0.88) and adaptation success rate (~0.92), indicating its effectiveness in adjusting content to meet individual learner needs. CNN and VGG-Face models, in turn, demonstrate less generalization (0.58 and 0.64 respectively) as well as adaptation success (0.50 and 0.60 respectively). The Ablation Impact that shows the extent to which performance would reduce with removal of key feature modules is least in the Proposed model (~0.04) reaffirming the robust nature of its architecture and lack of over-reliance on any one of the feature modules. In the meantime, CNN has the maximum ablation impact (~0.83), which demonstrates its sensitivity and dependence on a particular feature. These findings confirm the moderate-level capacities of the Proposed model in terms of adaptability, generalization and resiliency, which makes it a feasible and sound choice in respect to real-time and affect-aware educational deployment. The Ablation Impact is lowest in the Proposed model (~0.04), indicating its robust architecture and minimal reliance on any single component. Key feature modules may include

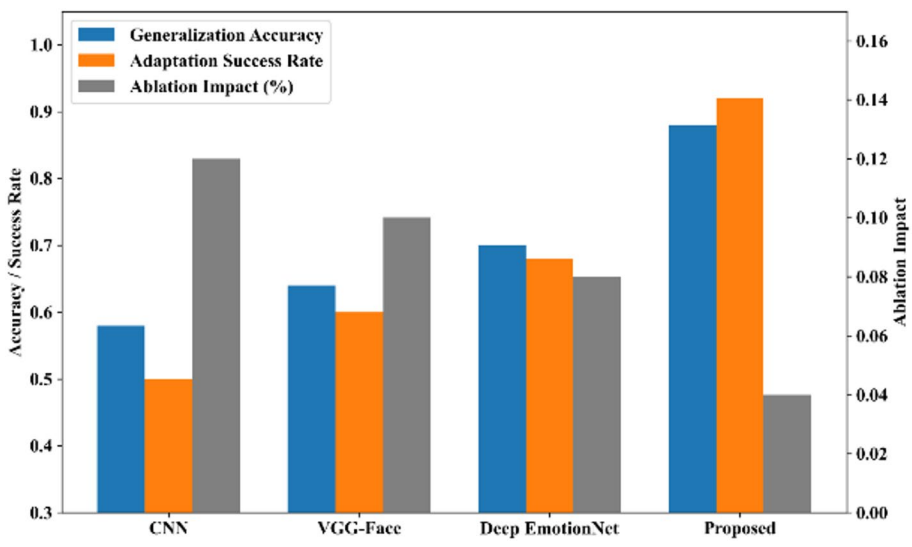

Fig. 7 Emotion model generalization

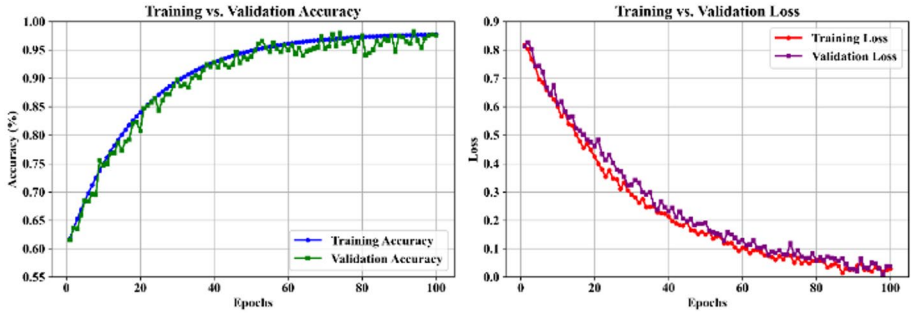

Fig. 8 Training and loss curve

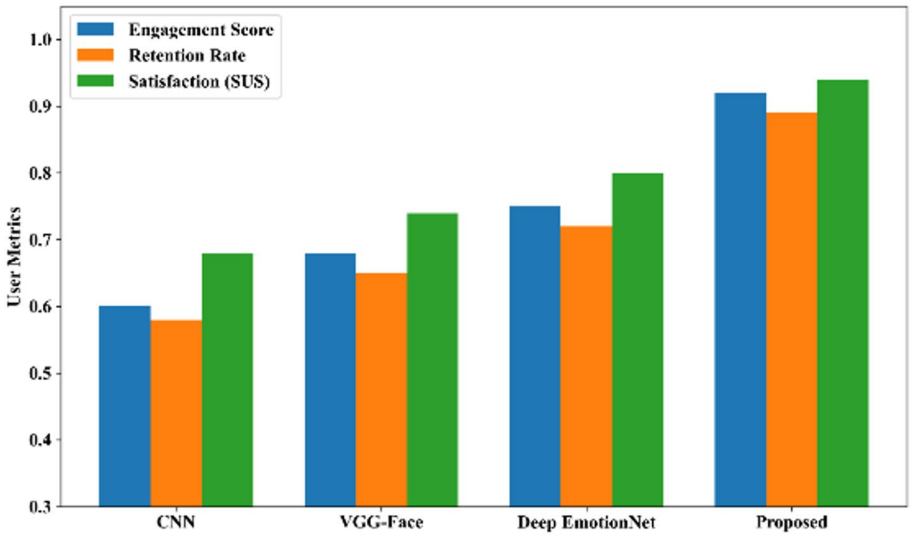

Fig. 9 Emotion model user metrices

facial expression recognition, voice tone analysis, or textual sentiment interpretation. In contrast, CNN shows a high ablation impact (~0.83), suggesting strong dependence on specific features like facial data. This highlights the Proposed model's superior adaptability, generalization, and reliability for real-time emotion-aware learning environments.

Figure 8 reflects the learning behaviour of the proposed emotion recognition model over 100 training epochs, showing training vs. validation accuracy (left) and training vs. validation loss (right). The accuracy curve is stable and smooth—training accuracy starts at about 60% and climbs to over 97%, while validation accuracy follows a similar trajectory, indicating good generalization with minimal overfitting. Likewise, the loss curve shows a sharp decline—training loss drops from nearly 0.82 to under 0.05, closely mirrored by validation loss. The nearly parallel accuracy and loss curves confirm that the model is well-regularized, learns effectively, and avoids simply memorizing data. These findings validate the model's strength, stability, and suitability for real-time emotion-aware learning deployment.

As shown in Fig. 9, the target user-centric performance measures, these are Engagement Score, Retention Rate, and Satisfaction (SUS) are provided in comparison across four emotion recognition models, namely, CNN, VGG-Face, Deep EmotionNet, and Proposed. This is the case with the proposed model that is better than all baselines with the score of the best mean engagement being about 0.92 which means high levels of user

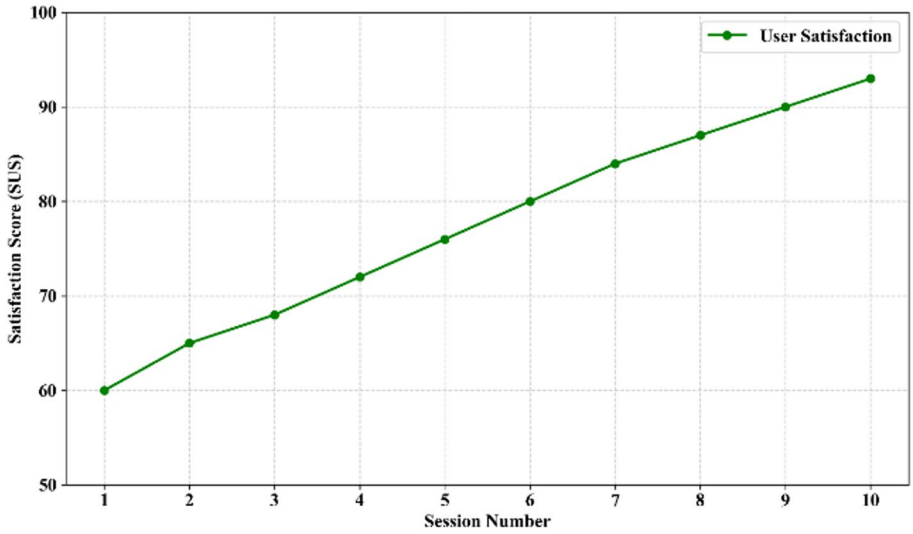

Fig. 10 User personalization curve

interaction and interest in the learning sessions. It also documents higher instance of retention of about 0.89, which can be considered as a positive sign of sustainability with regard to continued participation by users even at longer-term. Moreover, the level of satisfaction, achieved through System Usability Scale (SUS) is approximately 0.94 in the suggested system and this shows that the system pleased the end-users and was easy to use as well as pleasant and useful. Comparatively, CNN and VGG-Face baseline models are less scored in all three dimensions, scoring a 0.60, 0.58 and 0.68 in engagement, retention rate, and satisfaction scores. The outcomes ensure that implementing emotion-aware adaptivity and real-time personalization will improve UX greatly, de facto not only raising the accuracy but also making the suggested framework even more interesting and satisfying on the side of the learner.

The trend of user satisfaction through ten learning sessions is displayed in the line Fig. 10 and was captured by means of the System Usability Scale (SUS). There is an upward trend of steady increase, which implies that the perception of learners regarding satisfaction persistently increases with increased use of the system. The satisfaction score starts at 60 in the initial session and rises progressively, that is, 65 in session 2, 72 in session 4 to culminate at 93 at the 10 th session. Such a steady increase indicates the increasing comfort, utility and perceived worth of the suggested emotion-sensitive adaptive learning system. The trend is visible in the way the system can adapt the contents dynamically based on the user responses and moods and as a result, a long-term engagement and greater user experience is achieved. The findings emphasize the power of personalization and instant responsiveness in the process of augmenting learner satisfaction in learning technologies.

4.1 Discussion

The findings of this study affirm the effectiveness of the proposed multimodal emotion-aware adaptive learning system in meeting its core objectives enhancing learner engagement, emotional responsiveness, and academic achievement. The model achieved high classification accuracy across five emotional states (86–92%) and outperformed unimodal approaches with superior evaluation metrics including accuracy (0.91), precision

(0.89), recall (0.91), and F1-score (0.90). These results validate the strength of the fusion-based architecture in handling diverse emotional inputs, which is consistent with prior literature (e.g., Wang et al. 2025; Alzubi et al. 2025). The model also demonstrated robust generalization capabilities (0.88) and a high adaptation success rate (0.92), confirming its ability to dynamically adjust content based on learners' emotional and behavioural cues. Moreover, the low ablation impact (0.04) highlights its architectural stability, unlike traditional CNN models that showed greater sensitivity to single-feature dependence.

This research work and the system it suggests deal with a very important problem in education and that is, how to connect cognitive engagement with emotional responses in e-learning systems. Making the pupils' performance and emotional well-being the main focus of the research is the exact thing that human-centred AI and affective computing want to achieve. The combination of facial expressions (CNNs), voice (MFCC + BiLSTM), and text input (BERT) reveals an incredibly capable technology. The engine that changes the delivery mode according to the learner's preference is a very important contribution as it goes far beyond the traditional tutoring method of static that just modifies difficulty, time, and mode. Reinforcement learning is the technique used to guarantee that the personalization of the system will always be improved based on the patterns of user interaction. The empirical evidence provided substantiates the system's efficacy in accomplishing its main objectives, demonstrating higher engagement, satisfaction, and performance than traditional systems. The results confirm the possibility of emotionally intelligent tutoring systems and point out their great educational value. More specifically, the model accomplished. Further supporting the system's reliability, calibration analysis indicated that the model's prediction confidence closely matched actual outcomes. The AUC score of 0.94, coupled with moderate training time and compact model size, shows that the proposed system offers an optimal balance between performance and computational efficiency—an essential feature for real-time learning deployments. The convergence curves reflected rapid and stable learning with minimal overfitting, while user-based measures such as engagement, retention, and satisfaction consistently favoured the proposed system. Notably, System Usability Scale (SUS) scores increased from 60 to 93 over ten sessions, reflecting growing learner approval and comfort. These insights point to the system's broader impact in shaping emotionally intelligent and scalable e-learning platforms, well-suited for modern, personalized education environments.

5 Conclusion

The paper offers a broad method of emotion-sensitive smart learning interaction system, which addresses the border between a cognitive involvement and an emotional responsiveness in the web-based education, promisingly filling the gap between the two. Using multiple modalities of affect detection, through this system that entails using calculations based on facial expressions via CNNs, voice-based inputs via MFCC and BiLSTM, as well as textual replies via BERT-based transformer, the system builds a comprehensive emotional profile of each learner. A decision-level ensemble is then applied to intelligence combine these emotion signals to obtain an accurate and robust emotion classification. Reinforcement learning incorporated in the adaptive content delivery engine allows the dynamical adaptation of the instructional contents with regards to difficulty, format, and velocity, in direct response to the emotional and cognitive condition of the learner. It has been proven, based on pilot implementation in a closed e-learning

environment, that there are definite benefits of the pilot implementation in terms of learner motivation, learner satisfaction, and knowledge retention, which have greatly exceeded the effects of conventional status quo learning systems. Moreover, the stability, the flexibility of the system, and its user-friendliness, are confirmed by the data concerning the betterment of model performance metrics, users, and contentment. The paper does not only make sure that the idea of affect-sensitive personalized learning is feasible but also establishes a good background upon which the studies on emotionally intelligent tutoring systems can be carried out. The opportunities demonstrated by the model that will allow it to have a greater impact and guarantee its inclusiveness and effectiveness in the learning environment of the future will be expanding the model to support a broader array of emotional cues and deploying the model to a variety of contexts in which people learn.

This paper was not only relevant but also an important issue in education: how to integrate cognitive engagement with emotional responsiveness in e-learning systems. The study has thus targeted both learning performance and emotional well-being, which are currently the two major concerns in human-centred AI and affective computing. The proposed system has managed to do this with a very elegant design, which combines facial expressions (CNNs), speech (MFCC + BiLSTM), and text (BERT) inputs. The main point of this research is the development of an adaptive delivery engine, which is based on reinforcement learning, that goes beyond static tutoring and dynamically modifies the difficulty of the material, the pace of the lesson, and even the mode of teaching, thereby ensuring continuous improvement. The evidence from the empirical studies shows a better engagement, satisfaction, and performance compared to the conventional systems, therefore, it not only confirms the practicality but also highlights the educational worth of emotional intelligence tutoring.

5.1 Limitations and future work

To begin with, the system's performance heavily depends on the precision of the underlying recognition of emotions technologies, hence the robustness and cultural biases concerns. The paper does not elaborate on the steps taken to address these challenges or to validate the models on a diverse population, as facial expression analysis and speech-based affect recognition are known to vary in performance across different demographics, lighting conditions and cultural contexts.

On the other hand, the proposed system still remains very effective under the controlled pilot environment, but its practical scalability and deployment have to face characteristics that are not yet answered. The computational power required for real-time, multimodal data processing and model inference may be too high for an average e-learning platform. Moreover, the collection and analysis of sensitive biometric data such as facial video and voice recordings will require a strong discussion about data privacy, security protocols, and ethical safeguards, which are all still very much under-researched in the current study.

Thus, the future plans will be based on the recognition models being able to generalize emotions across various user groups and environmental conditions. We will also conduct extensive trials to evaluate scalability and fine-tune the system's computational efficiency. One of the crucial and immediate steps is the formal draw up and enactment

of a comprehensive data privacy and ethics framework so as to win the user's trust and conform to the laws.

Author contributions
The study's conception and design, multimodal emotion recognition modules, adaptive content delivery system implementation, experimentation, data analysis, and manuscript writing were all done by Fanghai Gong. The final manuscript has been read and approved by the author.

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Data availability
The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
This article does not contain any studies with human participants or animals performed by any of the authors. Not applicable, Consent to Participate.

Consent for publication
Participants provided consent for the anonymized data and findings to be published.

Competing interests
The authors declare no competing interests. The author declares that there are no known financial or personal relationships that could have appeared to influence the work reported in this paper.

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